

Anirudh Sivarajan (he/him)

La Jolla, CA • asivarajan@ucsd.edu • 630-605-2239 • www.linkedin.com/in/anirudh-sivarajan

EDUCATION

University of California San Diego

August 2025

Master of Science, Bioengineering (MS)

GPA: 3.67/4.0

Purdue University

December 2022

Bachelor of Science, Biomedical Engineering (BS)

GPA: 3.30/4.0

RESEARCH EXPERIENCE

Cognitive Science Laboratory • *Graduate Research Assistant*

La Jolla, CA | 04/2024-08/2025

- Designed a biologically-inspired recurrent neural network (RNN) in PyTorch, modeling individual neurons using first-order differential equations
- Mapped trained RNN connectivity onto a connectome-constrained Drosophila spiking neural network (SNN)
- Revealed circuit-level mechanisms underlying persistent activity during memory encoding, contributing to a first-author manuscript and SFN abstract submission

Microbiome Science Laboratory • *Graduate Research Assistant*

La Jolla, CA | 08/2023-04/2024

- Processed and analyzed large-scale metagenomics data (16S rRNA) using QIITA, QIIME2, and Python/R to identify microbiome-drug interactions
- Evaluated how microbiome alterations influence Alzheimer's disease (AD) progression and associated neuropathology
- Identified microbial signatures linked to neuroinflammation, supporting the hypotheses on gut-brain axis in AD pathology

Functional Genomics Laboratory • *Research Assistant*

Evanston, IL | 03/2022-08/2022

- Used single-cell RNA sequencing to study how gene expression influences kidney development in mice
- Improved clustering accuracy by 15% with Seurat, enhancing cell-type resolution and enabling novel marker identification
- Reduced batch effect noise by 20% by applying unsupervised clustering to QC metrics

Host-Microbiome Interactions Laboratory • *Research Assistant*

West Lafayette, IN | 12/2020-12/2022

- Operated different dimensionality reduction algorithms such as PCA & PCR to translate drug responses
- Trained regression models to help translate the gene expression in a clinical dataset to different human responses
- Gained an understanding of the sleep-wake cycle's effect on the progression of Alzheimer's disease

WORK EXPERIENCE

Mathematical Methods • *Teaching Assistant*

La Jolla, CA | 10/2024-12/2024

- Graded homework assignments for 50 students on a weekly basis
- Answered questions and conducted weekly office hours

Phase Maze LLC • *Mechanical Engineer*

Naperville, IL | 07/2021-07/2023

- Manufactured different components of the toy, a stackable maze, and effectively reduced the turnaround time by 50%
- Identified failure points in the stacking mechanism and iterated on design using 3D printing and physical prototyping
- Drafted detailed reports documenting design revisions and mechanical performance to support engineering decisions

ADDITIONAL SKILLS

Certifications: Applications in Data Science, 2019

Programming & Data Analysis: MATLAB, Python, R, SQL, NumPy, Pandas, Scikit-learn

Machine Learning & Deep Learning: TensorFlow, PyTorch

Data Visualization: Matplotlib, Seaborn, Tableau, Power BI